

Nirmal Gopalakrishnan

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PROFILE

Computer Science Engineering graduate (2026) with hands-on experience in Python, Java, deep learning, data analytics, and game development. Demonstrated ability to design and deploy end-to-end ML systems including CNN-BiLSTM models and AI-powered recommendation engines. Built a multi-page sales analytics dashboard in Power BI during a data analytics internship, applying data cleaning, DAX, and business intelligence skills on real-world retail data. Proficient with tools such as TensorFlow, Keras, Flask, SQL, Power BI, and Excel. Quick learner with a strong foundation in OOP, cloud computing, and secure design principles.

PROFESSIONAL EXPERIENCE

Clover Technology, *Game Developer Intern* Aug 2025 – Jan 2026 | Coimbatore

- Trained in Blender and Unreal Engine to build 3D assets and game-ready environments.
- Gained practical experience in 3D modeling, texturing, and animation workflows.
- Integrated optimized 3D assets into game engines for visually engaging gameplay experiences.

IPCS Global Pvt Ltd, *Data Analytics Intern* Jun 2025 – Jul 2025 | Coimbatore

- Performed data cleaning, analysis, and visualization using Python, SQL, Excel, and Power BI.
- Developed analytical skills through real-world datasets and reporting tasks.

EDUCATION

Kathir College of Engineering 2022 – 2026 | Coimbatore

Bachelor of Engineering, Computer Science and Engineering
CGPA: 7.1

Gandhi Matric Higher Secondary School, HSC 2021 – 2022 | Namakkal

Percentage: 65%

Gandhi Matric Higher Secondary School, SSLC 2019 – 2020 | Namakkal

Percentage: 89%

PROJECTS

Deep Learning-Based Stuttering Detection in Speech Signals Apr 2026

- Designed a hybrid CNN-BiLSTM model to detect stuttering patterns such as repetitions, prolongations, and speech blocks.
- Applied MFCC-based feature extraction to capture acoustic characteristics of speech signals.
- Integrated ASR-based speech-to-text processing using OpenAI Whisper for audio transcription.
- Developed an NLP pipeline using spaCy and NLTK to convert disfluent text into fluent sentences.
- Implemented text-to-speech (TTS) output and built an interactive diagnostic dashboard for analysis.
- Technologies: Python, TensorFlow, Keras, Librosa, OpenAI Whisper, spaCy, NLTK, Flask

Sales Performance Dashboard - Power BI Jul 2025

- Designed and built a 2-page interactive sales analytics dashboard in Power BI for Balaji Fast Food, covering transaction and inventory metrics.

- Page 1 features a Pie Chart, Line and Stacked Column Combo Chart, Area Chart, KPI Card, Donut Chart, and a Slicer for dynamic filtering by staff (received_by field).
- Page 2 includes a Treemap, Pivot Table, and Multi-Row Card to analyze quantity sold and revenue by item type and time of sale.
- Performed data cleaning and transformation on sales data fields including item_name, item_type, item_price, quantity, transaction_amount, transaction_type, and time_of_sale.
- Technologies: Power BI, Data Cleaning, Data Visualization, DAX, Business Intelligence, Excel

AI-Powered Personalized Recommendation Engine

Apr 2025

- Built a recommendation engine using collaborative filtering, content-based filtering, and a hybrid model.
- Achieved a Mean Squared Error (MSE) of 0.89 with 80% positive user satisfaction from feedback.
- Deployed via Flask web API with MySQL and MongoDB for data storage.
- Technologies: Python, TensorFlow, PyTorch, Surprise library, Flask, MySQL, MongoDB

Automated Teller Machine System

Jan 2025

- Engineered a Java-based ATM system applying OOP principles including abstraction and encapsulation.
- Implemented core features: balance inquiry, cash withdrawal, and PIN change with secure authentication.
- Conducted unit testing using ATMSimulation to validate functionality.
- Technologies: Java, OOP, Secure Design, Software Testing

CERTIFICATIONS

Introduction to AI and Vector Search (Dec 2023) | **TensorFlow 2.0 for Deep Learning - Infosys** (Dec 2025) | **Introduction to Neural Networks - SimpliLearn** (Dec 2025) |

Oracle Cloud Infrastructure - Oracle | **Java Programming - Smart Yugam Academy** (Dec 2025) | **Database Management Systems (DBMS) - Infosys**

CORE SKILLS

Programming Languages (Python, Java, C) | **Machine Learning / AI** (TensorFlow, Keras, PyTorch, CNN, LSTM, NLP (spaCy, NLTK), OpenAI Whisper) | **Data Analytics** (Python, SQL, Excel, Power BI, DAX, Data Cleaning, Data Visualization, Business Intelligence) | **Game Development** (Blender, Unreal Engine, MetaHuman, 3D Modeling, Texturing, Animation) | **Web and Backend** (Flask, REST API, MySQL, MongoDB) | **Concepts** (OOP, Data Structures, Algorithms, Secure Design, Cloud Computing)

LANGUAGES

English (Professional Proficiency) | **Tamil** (Native)